

TME

poste 2.

$$X = \frac{a}{b} [P]$$

$$\Leftrightarrow b X = a [P]$$

$$\begin{array}{ccc} \uparrow & \uparrow & \uparrow \\ x_1 & a & x_2 \end{array}$$

$\Leftrightarrow$ système de $\nearrow$ $\begin{matrix} \text{variables} \\ \text{count.} \end{matrix}$

$$\begin{pmatrix} 1 & a \\ 0 & P \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad \begin{array}{l} x_1 = b \\ x_2 = 1. \end{array}$$

post 3.

$$p \equiv 1[4], p = a^2 + b^2?$$

$$a^2 + b^2 \equiv 0[p]$$

$$\Leftrightarrow a^2 \equiv -b^2[p]$$

$$\Leftrightarrow a^2 = (\alpha b)^2 \pmod p \quad \text{avec} \quad \alpha^2 \equiv -1[p]$$

$$\Rightarrow a = \pm \alpha b[p]$$

$$\Rightarrow (b, k) \begin{pmatrix} \alpha & 1 \\ p & 0 \end{pmatrix} \begin{matrix} \text{vector} \\ \text{can't} \\ \text{be} \end{matrix} \begin{matrix} \swarrow \\ \searrow \end{matrix} \underline{\underline{= (a, b)}}$$

Trouver α t.g. $\alpha^2 \equiv -1[p]$.